

Pababy: An Interactive System for First-time Parents to Learn Neonatal Nursing

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ABSTRACT

Learning neonatal nursing is an important task for parents. Early and appropriate neonatal nursing education can not only ensure the healthy development of newborns, but also help establish early safe dependence between parents and newborns. In this paper, we introduce Pababy, an interactive system which could support first-time parents to effectively master basic nursing knowledge of newborns before delivery through practicing. Pababy consists of a simulator and an application. During the practice process, the simulator and application will give real-time feedback according to the action of parents. Besides, the application will also display step-by-step teaching while parents' practicing. We also conducted a preliminary user study of Pababy. The result shows that parents have a high acceptance of Pababy while learning and gain more confidence to participate in nursing.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design process and methods;

KEYWORDS

Nursing learning, Simulation, Sensing technology, Prenatal, Neonate, Parent

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1 INTRODUCTION

It is a challenge for first-time parents to take good care of neonates especially in the first 4 weeks, during which they are under physical and psychological pressure.

First-time parents often lack necessary knowledge of newborn care without appropriate prenatal training, which leads to a large number of wrong actions in the postpartum neonatal nursing [26], causing serious harm to the health of neonates and aggravating parents' anxiety [19]. Similarly, due to low experience, first-time parents may lack emotional preparation for the arrival of newborns, which is not conducive to the establishment of early emotional relationships [11]. Despite considerable efforts invested in helping parents take care of newborns, many previous works focused on infant's monitoring and warned only in sudden and unexpected times of crisis [24], placed education in the postpartum stage and ignored the time and patience required for parents to internalize nursing knowledge [9].

The interaction between people and multi-sensor simulator has attracted extensive attention of researchers of HCI. Ledesma et al. [22] proposed a traumatic brain injury simulator that could be used to educate students and confirmed that the simulator could reduce the time to learn clinical skills in the field of medical education [18, 22], and warned if they made mistakes [16, 22]. Kunimura et al. [13] proposed Yotaro, a baby-shaped simulator that invited users to interact with it. Yotaro could create various behavioral feedbacks during conversation, and it had been proved to increase the user's mood and aid in the development of emotional connection. Our

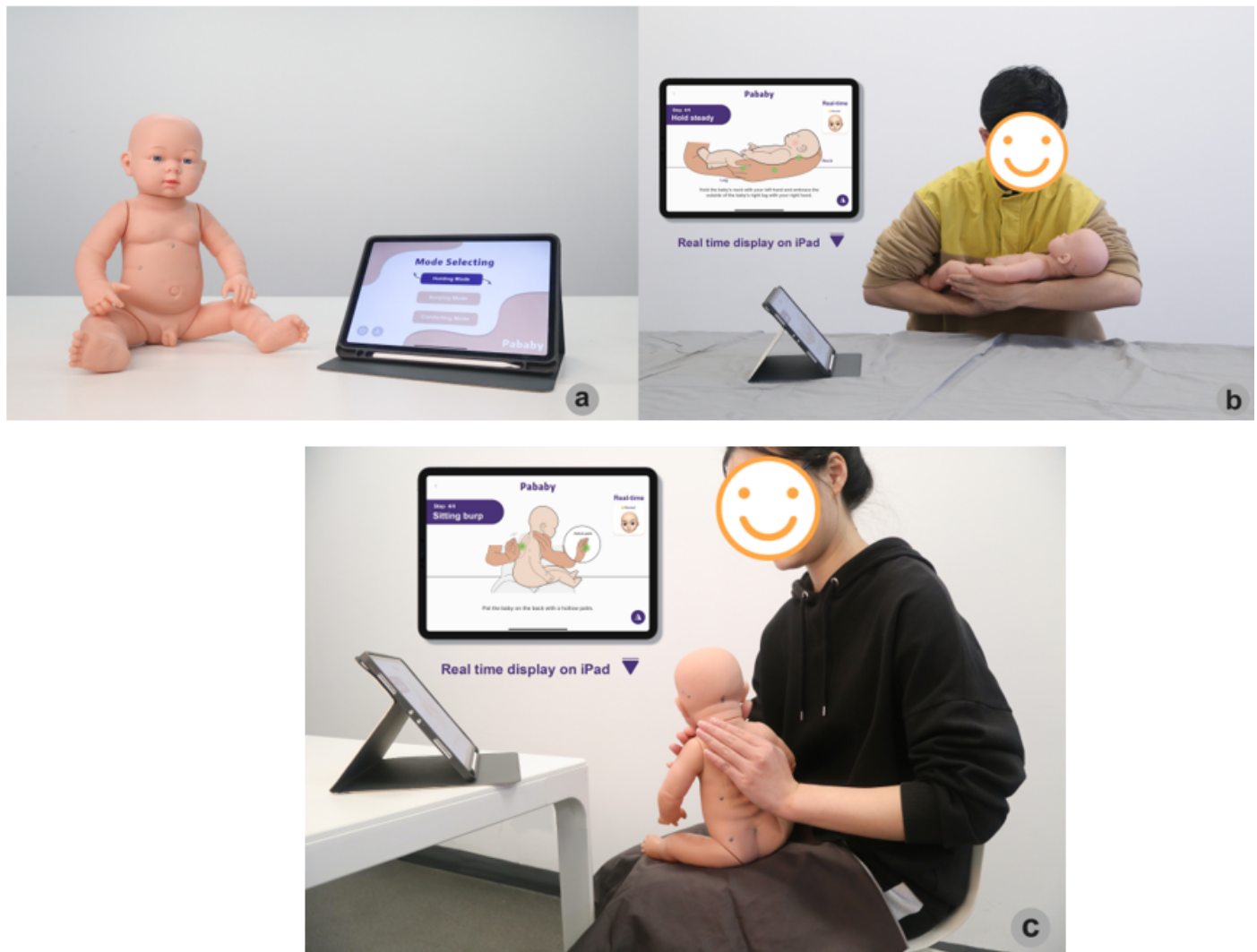


Figure 1: (a) Pababy System (b) (c) Pababy supports parents while neonatal nursing learning

work exploited the previous definition of simulator but combined it with step-by-step teaching and real-time feedback to support parents while nursing learning.

In this paper, we introduce Pababy, an interactive system for first-time parent (Figure 1), including a simulator and an application, based on sensing technology. When interacting with Pababy, users learn and practice neonatal nursing through step-by-step teaching on the application and the simulator judges whether the parents' action is correct according to the sensor signal at the same time. Pababy will remind users by giving correction suggestions if they performed wrong actions.

To build Pababy, we conducted an education probe with experts and developed the prototype. In order to evaluate parents' views of Pababy, we conducted a preliminary user study. The result showed that parents were willing to use Pababy in learning process and

found the real-time reminder useful while practicing. Similarly, parents gained more confidence and looked forward to the arrival of newborns after using Pababy. The main contributions of our work are: (1) We designed an interactive system, including a baby simulator and an application, to strengthen parents' practice in form of real-time feedback while nursing learning. (2) We presented a method for detecting incorrect behaviors throughout parents' learning and practice of neonate nursing and displayed parents' learning process. (3) We further derived support method of neonatal nursing learning through prototype and semi-structured interviews.

2 RELATED WORK

2.1 Neonatal nursing support

Prior works confirmed that neonatal nursing support tools could effectively improve the quality of neonatal care [7, 9, 23] and reduce the burden of parenting [2, 24]. Due to this importance, numerous

neonatal nursing supporting methods have been presented [7, 20]. For example, baby lucent [9] aimed to help first-time parents reflect upon and improve their parenting behavior based on real data. Modhera [25] is a modular toolkit for parents to take care of their babies by recording and monitoring baby physiology data in different scenarios. Wei et al. [4] used flexible materials to develop a wearable sensor system, providing accurate as well as comfortable temperature monitoring for neonates. However, prior works mostly focused on the neonatal postpartum state [25] by monitoring the data of neonates and environment [18], which alerted only when neonates were in a critical circumstance. Besides, they were weak in providing universal nursing for parents and neglected the time and patience needed for parents to learn and master. Alena et al. [17] focused on parents' daily feeding, designed a VR training system and verified that multiple simulation exercises can improve the feeding efficiency of first-time parents [17]. Inspired by this, in this paper we explore the possibility of parents learning neonates nursing knowledge before birth, so as to support parents to take care of neonates more safely.

2.2 Baby simulator

Simulation contains a wide range of technologies aimed to recreate aspects of the real world [6]. From educational and medical perspective [8, 16, 21], simulation may assist students in developing abilities [21] in scenarios that are comparable to those seen in real life while also reducing the danger caused by inappropriate operation [6, 18]. Sensing technology has become more and more mature and now has the potential to realize baby simulators [13]. For example, Fatihah et al. [7] developed an infant simulator with Philips hue led kit for Cyanosis Baby Manikin Application with the Philips Hue LED Kit for medical use. Mochizuki et al. [15] found that compared with virtual technology, tangible baby could make parents feel the happiness of nursing neonates, reduce parents' anxiety [9, 15], and was more conducive to the establishment of emotional connection between parents and newborns [15, 18]. At the same time, he proposed the possibility of baby simulator in home nursing [13, 15]. However, the current research on family baby simulators mostly encourages parents to experience [10] and lacks clear and efficient guideline on how to operate properly [17]. On the basis of previous work, we further explore the benefits of baby simulator with sensor technology that provides real-time feedback and step-by-step teaching to assist parents acquire daily nursing skills and accumulate emotional experience.

3 DESIGN PROCESS

To design and build Pababy, we firstly conducted design probe sessions with experts. Here, we describe the design process and the resulting design goals.

3.1 Design probe with experts

We conducted a preliminary study with 3 experts (2F, 1M, mean age=36, SD=6.68, P1, P2 and P3) from maternal-infant center in Dalian, China, who have more than 3 years of neonatal nursing experience and have conducted relevant education. Every design probe session lasted roughly 60 minutes, including an introduction of our preliminary ideas, a discussion and communication

with the experts sharing their previous nursing experience and a semi-structured interview with their vision of our prototype. They all agreed with our starting point that "prenatal neonatal nursing education is really necessary, and it will be more efficient through practice" and made suggestions as follows: (i) P1 and P2 recommended that timely error correction feedback is necessary in neonatal nursing learning. (ii) P1 and P3 revealed that although many parents had carried out nursing learning, due to the lack of confidence, the neonates' casual crying would make parents feel overwhelmed in the real situation. (iii) P3 suggested that we give teaching step by step to reduce parents' learning cost. (iv) P2 proposed that in nursing education, direct physical contact with neonates is a difficulty of teaching, because parents often have no opportunity to practice before birth.

3.2 Design goals

Informed by our design probe sessions with experts and relevant prior work, we synthesized the following key goals for an interactive system to support first-time parents to learn neonatal nursing: (1) Real-time feedback during practice to avoid constant incorrect practice and lay little stress on learning tasks. As ((i), in 3.1) shows timely error correction may be a necessary way for guiding parents. (2) More efficient practices through simulation exercises, which not only supports parents' learning, but also help build parents' nursing confidence. (3) Consider the direct physical interaction between parents and neonates as pointed out in ((ii) in 3.1), which is an important issue since it deals with the realism of the baby simulator and its behavior.

4 SYSTEM DESCRIPTION

Based on the design probe with experts, we developed the prototype, Pababy. Pababy is comprised of hardware and software (Figure 2). The hardware refers to a simulator with functions of vibration amplitude detection, force point detection and body inclination angle detection. The software is an application that provides step-by-step teaching and gives real-time feedback. Software and hardware are connected via Wi-Fi.

4.1 Application

Parents can use the application (app) to select the simulator's mode and start practicing (Figure 3a). Furthermore, the app works in conjunction with the simulator and shows step-by-step instructions while training (Figure 3b). Meanwhile, when parents are practicing, the app will display real-time emotional expressions and pop-up reminders based on parents' actions (Figure 3c). For example, if the simulator identifies a fault action from parents, it will send a signal to the application, where a reminder of the problem and corrective suggestions will be displayed in real time.

4.2 Simulator

The simulator of Pababy is made of silica gel totally and integrates different sensors and actuators to simulate function of multiple parts (Figure 4). In order to increase the sense of reality, we add counterweight in the simulator, which is commensurate with the 2 weeks neonatal average weight of 4kg. We use Arduino Mega as development board, which can collect and transmit sensor data.



Figure 2: Conceptual framework of Pababy



Figure 3: Interface of Pababy APP (a) Mode selection (b) Step by step teaching (c) Error reminder and suggestions.

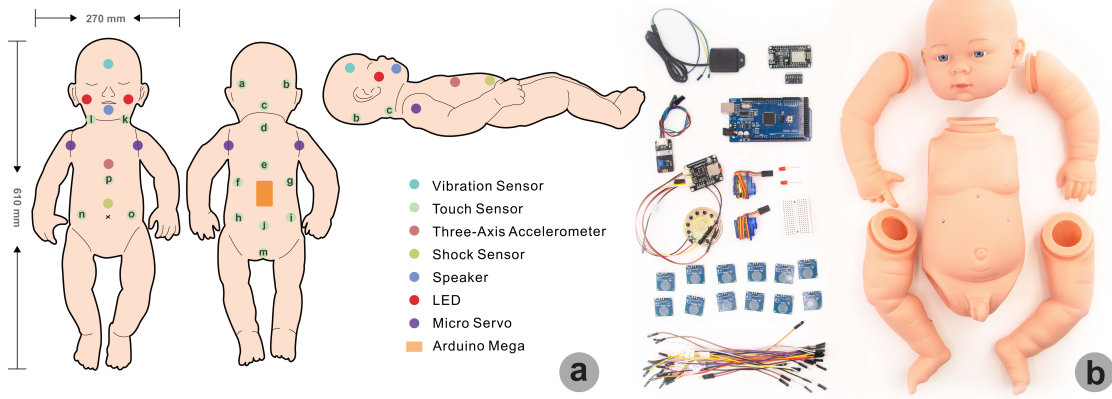


Figure 4: The simulator of Pababy (a) Sensor layout of simulator (b) Composition of simulator

Arduino Mega sends a digital switch signal to the back-end server through the Wi-Fi module (esp8266), and the front-end page on the iPad obtains the digital switch signal on the back-end server through HTTP request. At the same time, actuators and the application begin to work.

As stress detection in the simulator, we employ 16 ttp223 touch sensors in the shape of a dot matrix. We put solder balls on the simulator skin as contact points and ensure that different contact points don't interfere with one other. When user touches the contact point, the output level status is switched. We can judge whether the user touches the contact point and whether the user's posture is right based on the high or low level.

As the body tilt angle detection, we use an ADXL345 three axis accelerometer and build it into the abdominal cavity of the simulator for it can track the motion state of objects. At the same time, ADXL345 three axis accelerometer can detect the static gravity acceleration and measure the instantaneous acceleration caused by the impact process. It is also used for vibration amplitude detection

together with shock sensor (yousee) and sen0433 vibration sensor (DF robot).

To indicate pain, we employ two types of baby's weeping and micro servo-controlled arm swings. We use two types of laughter, a type of hiccup and the red LED under cheeks to convey baby's comfort. We collect several voices of babies expressing their emotions in real scenes through online media, such as crying, happiness, discomfort and hiccups. All sounds are played by voicebox MP3 TF module (YW robot).

4.3 Interaction and principle

Based on the design probe, we chose to develop neonate holding, neonate burping and neonate comforting as three modes on the early prototype.

4.3.1 Holding mode. In holding mode, parents learn basic neonate holding posture (Figure 5). Neonate's neck is so weak that it can't fully support the heavy head [3]. When holding neonate, we focus on monitoring parents' support for neonate's head and neck. At

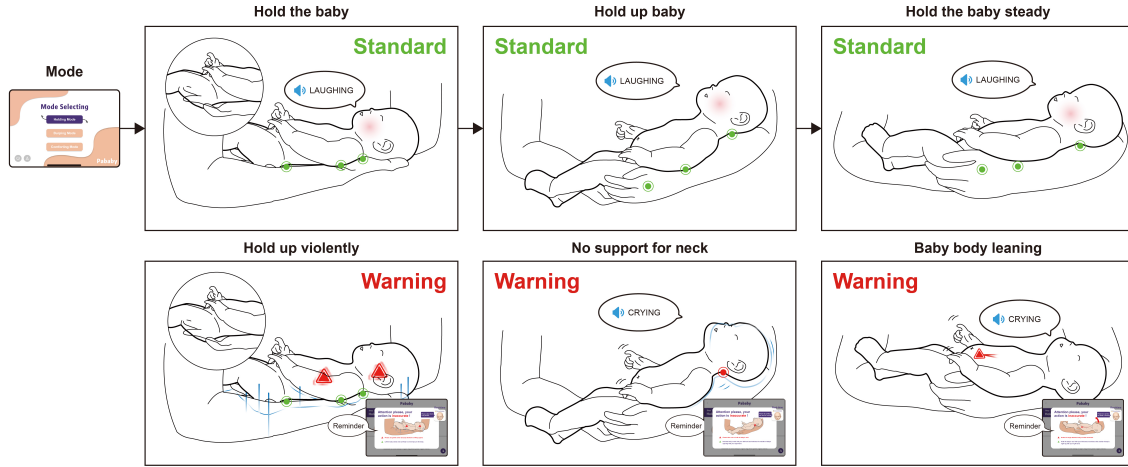


Figure 5: An example working diagram of holding mode.

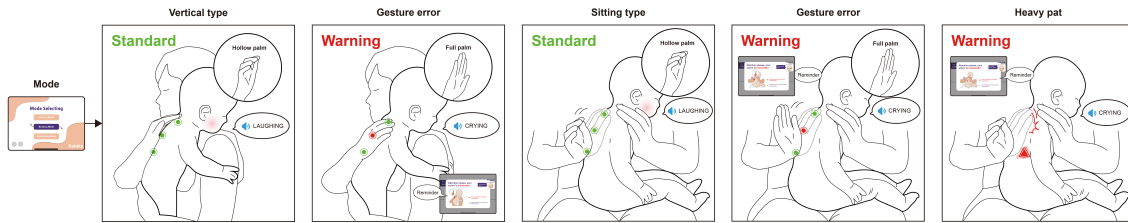


Figure 6: An example working diagram of the two types of burping.

the same time, Arduino mega board judges whether the parents' action is standard according to the input signal of the corresponding contact point on the skin. For example, when parent holds up the neonate, if there is a continuous signal output at touch sensors c, e and j, Arduino Mega collects and transmits data, and judges that it is now the in correct state. Then Pababy gives feedback of "comfort" and parents go to the next step. At the same time, ADXL345 three axis accelerator monitors the inclination between neonate's trunk and horizontal. If exceeding range of 5-55°, it means the head gives excessive pressure to the neck [3] and the baby simulator sends out an uncomfortable sound and reminder and correction suggestions will pop-up on the app in real time. In order to avoid neck injury caused by holding too fast or too violent, the vibration neck will monitor the head amplitude [1] and give a reminder during the process.

4.3.2 Burping mode. The burping mode includes 2 types: a) vertical type b) sitting type (Figure 6). In addition to correct posture and flapping position detection, it is particularly necessary to pay attention to the control of strength when burping, because excessive flapping strength may cause damage to the spine of new-borns [12]. We use shock sensor (yourcee) as trunk vibration amplitude monitoring and adjust the sensitivity threshold according to the appropriate strength guided by professional nurses. At the same time, through the output signal of the baby's central back touch sensor, we can judge whether the parent's gesture is the correct hollow tapping state.

4.3.3 Comforting mode. In the comforting mode, parents learn how to comfort neonates to sleep through hugging and patting (Figure 7). Patting and hugging are two common ways to help neonates feel relaxed. In this process, in addition to paying attention to the correct holding method, it is necessary to avoid excessive clapping and shaking [14]. According to the research of Maruki Development [5], the optimal frequency for infants is 0.55 Hz, for 0.55 Hz is similar to the mother's heart rate 67 BPM [18]. Besides, according to a study by Ying Hao, Ko et al [12], a neonate weighing 4 kg would be very uncomfortable when the manual clapping value of the cradle reached 1.3718 m/s². The ADXL345 three-axis accelerometer measures the shaking force and notifies with crying and swing arms if the Accelerometer reaches the preset level. When the parent's action stays correct, "baby" falls asleep with cheeks showing the effect of red breathing lamp.

5 PRELIMINARY USER STUDY

To confirm how parents interact with Pababy and their views of Pababy, we conducted a preliminary user study. We recruited 4 parents A, B, C, D from local neighborhood (2F, 2M, mean age = 26.5, SD=1.12, SD=0.74). All participants (pA, pB, pC, pD) have learned basic neonatal nursing through Internet or curriculums and the whole process was recorded with consent of the participants.

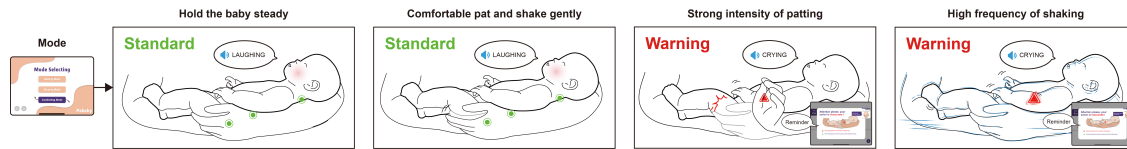


Figure 7: An example working diagram of comforting mode.

5.1 Procedure

The experiments were carried out in a classroom. First, we gave a short introduction of the prototype and how to use the system. Then, the participants were given 1 hour to independently train and learn by using Pababy. During this 1 hour, the participants were observed. When this was done, we encouraged participants to take Pababy home and continued using for 3 days. After using session, we explored participants' perceptions with a Likert scale questionnaire (Table 1) and conducted semi-structured interviews focus on three topics: (i) Their relationship with Pababy. (ii) Opinions on future development of Pababy. (iii) Difference between traditional learning ways and Pababy.

5.2 Results

As showed in Table 1 (a seven point Likert scale), all the participants showed a high willingness to use Pababy. However, they proposed different strategies for the step-by-step teaching. For example, pC, who scored 7 of Q (ii), told us that step-by-step teaching could make him master some activities more thoroughly, such as "hollow palm". pB, who scored 3 of Q (ii), didn't think it necessary to disassemble the teaching steps. While all four participants pointed out that the real-time feedback was helpful during practicing (an average score 6.75).

When participants were asked about their relationship with Pababy, pA smiled and said, "I think he is like my upcoming baby. I was a little nervous and afraid of making mistakes at the beginning, but with practice gradually I felt great." pC said: "I think he is much like my private tutor. I can practice whenever I want and get timely guidance when I make mistakes."

Similarly, participants looked forward to the follow-up work of Pababy and also gave us suggestions. pA and pB both expected that Pababy could have more learning modes in the future, such as feeding and changing diapers. pC said: "I don't want Pababy to be useless after birth if only it has other functions at that time." After a while, pC added, "for example, after the baby is born, it can become a toy to amuse the child, or a player to play music."

For the difference, pD said, "I think the previous study was more like understanding general knowledge, but Pababy lets me experience neonatal nursing truly and meaningfully." pB suggested that "The timely feedback is really great. In the past, I didn't know whether my learning online was correct or not."

We also found some interesting things during experiment. pC showed dependence on the app during using. She would habitually see the app every time she made an action, whether it is correct or not. In addition to practicing correct movements, pD would also try some inaccurate postures to see the feedback reminder of the system. pD said "If I understand the impact of wrong action on my baby, I will be more careful in future care."

In general, all participants are willing to use Pababy while neonatal nursing learning and our findings reveal Pababy as a significant partner to support parents to gain neonatal nursing knowledge efficiently and enhance confidence in neonatal nursing.

6 LIMITATIONS AND FUTURE WORK

We identified certain limitations in our research. Firstly, limited by the epidemic situation, we only found three experts as representatives of neonatal nursing knowledge. With more experts we might have gathered additional factors and aspects of importance. Secondly, the experiment's duration is not long enough for the system to develop with the parents' feedback and we may lack long-term company and observation of Pababy.

In the future, we will further improve the function of the system and expand the neonatal nursing coverage of Pababy. In addition, we will optimize the performance form and physiological index simulation of the system and apply more "micro feedback" to Pababy, making users feel more real. Considering that Pababy will carry more functional components, it is anticipated to further reduce the size of components in future design. Similarly, based on the suggestions from the experiment, we will also think about the role of Pababy in family after the arrival of the new-born. Finally, we will also try to recruit more users.

7 CONCLUSION

In this paper, we introduce Pababy, an interactive system which could support first-time parents in efficiently mastering basic nursing skills before delivery. Pababy provides real-time feedback and correction based on parents' action, allowing them to alter their actions in real time while practicing. Our user study showed that Pababy plays an active role in parents' understanding of neonatal nursing. Parents can learn and master neonatal nursing more efficiently by practicing with simulator, build confidence in caring for newborns and welcome the arrival of "new life".

Compared with prior work, our prototype was based on real-time feedback and error correction reminder, supporting parents to recognize inaccurate action and adjust in time, which may expand the research in the field of neonatal nursing supporting. Besides, we believe that Pababy can play an active role in more home nursing activities caused by geographical distance or infectious situations such as covid-19.

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Table 1: Results of parents' questionnaire

Questions	Male A	Male B	Female C	Female D	Average
i. Are you willing to use Pababy while neonatal nursing learning? (1: unwilling – 7: willing)	7	6	6	7	6.5
ii. Is it easy for you to understand how to use Pababy? (1: difficult – 7: easy)	7	6	6	6	6.25
iii. Is the step-by-step teaching of Pababy helpful? (1: useless – 7: helpful)	5	3	7	6	5.5
iv. What do you think of Pababy's real-time feedback and error correction reminder when you make an incorrect action? (1: useless – 7: helpful)	7	6	7	7	6.75
v. Do you have confidence to take care of the newborn safely after practicing with Pababy? (1: unconfident – 7: confident)	6	5	6	7	6

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